

White Paper

Authors

Nils Grevenbrock

Zhengpu Zhao

Nihil Patel

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Model Risk Management in the Age of AI

Abstract

Model Risk Management (MRM) must evolve from periodic validation to continuous governance as Generative and Agentic Artificial Intelligence (AI) systems introduce opacity, stochasticity, and autonomy that break traditional supervisory assumptions. This paper outlines a governance-centric MRM paradigm, proposing practical controls—such as dynamic guardrails, real-time monitoring, Human-in-the-loop (HITL) oversight, and third-party risk mitigation—to ensure safe, compliant, and resilient deployment of AI systems for regulated financial domains.

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1 Introduction

Traditional Model Risk Management (MRM) frameworks were built for deterministic, transparent models under full control. Supervisory frameworks such as the Supervisory Regulation 11-7 (SR 11-7), introduced by the Federal Reserve and the Office of the Comptroller of the Currency in 2011 assumed models with defined inputs, predictable outputs¹, and clear interpretability².

These conditions enabled static validation cycles and well-understood governance practices.

Generative AI (GenAI) and agentic AI systems break these assumptions by introducing opacity, stochasticity, and autonomy. The large language models (LLMs) internal structure functions as a black box, offering limited interpretability, producing nondeterministic outputs, and relying on a third-party infrastructure that might change without notice. Agentic systems further increase complexity by enabling autonomous decision-making under uncertainty.

Static validation is no longer sufficient; MRM must shift to continuous governance. While supervisory frameworks (such as SR 11-7) still stand at a high level, their application must evolve. Institutions need dynamic guardrails instead of static thresholds, real-time observability rather than periodic checks, and human-in-the-loop oversight for high stake decisions.

Managing third-party risk and monitoring the entire system's outputs and behaviors must become core components of governance. Monitoring models to approved versions and maintaining fallback models can mitigate the risk of unexpected vendor changes. Continuous, system-level monitoring through behavioral testing and anomaly detection ensures that the AI system remains aligned with intended objectives and provides resilience against performance drift or unintended behaviors. Also, monitoring must extend beyond individual model components to the entire system, focusing on monitoring inputs, outputs, and emergent behaviors rather than internal mechanics alone.

This paper first traces model evolution across three generations, then proposes operational strategies for model risk management, including a case study. Finally, the paper concludes with actionable steps institutions should consider.

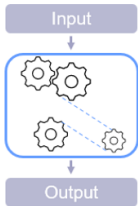
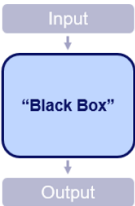
¹Traditional models yield deterministic estimates, whereas generative AI models (for example, LLM) sample from learned distributions, introducing intrinsic stochasticity.

²Moody's adheres to Pérez-Cruz et al. (BIS, 2025), distinguishing interpretability (understanding model mechanics) from explainability (justifying outputs), though terms are often used interchangeably.

2 The Evolution of Models: From Traditional Models to AI Systems

Models have evolved through three generations, trading transparency, determinism, and control for performance and capability.³ Each step, from traditional scorecards and structural models to Machine Learning (ML) models to GenAI-based systems, introduces new risks that challenge conventional validation frameworks. Table 1 summarizes how risks and mitigation strategies evolve across generations.

Table 1: The Evolution of Models in the Age of AI

	Gen 1: Traditional Models <i>1960s - 2000s (dominant)</i>	Gen 2: Machine Learning <i>2000s - 2020 (explosive growth)</i>	Gen 3: AI Systems <i>2022 - present (emerging)</i>
			
Examples	• Scorecards • Linear models • Structural models	• Tree-based models • Kernel-based models • Small/medium neural networks	• LLM-assisted tools • LLM-embedded workflows • Autonomous LLM agents
Capabilities	• Mathematical formulation of understood relationships • Best for well-specified, structured data problems	• Letting data reveal empirical patterns not easily specified • Superior on high-dimensional, data-driven tasks	• Operate over unstructured and multimodal data (text, video, image, audio) at scale • Supports reasoning, planning, acting, learning, evaluating, tool use, and orchestration
Transparency	●	●	●
Determinism	●	●	●
Control	●	●	●

Generation 1: Traditional models are deterministic, transparent, and fully controlled. These models, such as logistic regressions, rely on well-understood relationships and structured data. Inputs and outputs are predictable. Risks, such as model misspecification or parameter instability, are traceable and manageable through established MRM techniques.

EXAMPLE

Historical data shows a company's financial leverage ratio correlates with a company's credit risk. This relationship can be modeled via logistic regression. The estimated coefficient of this logistic regression can be observed and interpreted. From this we can derive statements such as "If the financial leverage ratio of a company goes up by x%, on average, the credit risk increases by y%."

Generation 2: Machine Learning models deliver higher predictive power but reduce interpretability. Techniques like Random Forest can uncover complex patterns in the data beyond what could be encapsulated in traditional models but introduce opacity due to the possibly vast number of parameters and complex non-linearities. In these situations, it is almost impossible to connect model outputs back to parsimonious human-understandable rules operating on inputs. Meanwhile, ML models remain as deterministic functions that modelers typically control and maintain,⁴ much like traditional models.

EXAMPLE

A Random Forest model can be trained to find complex patterns in credit default data that traditional linear, or structural (non-linear) models cannot capture. How the model arrives at a prediction, for example, a combination of firms' financial leverage and firm size contributing to the increase in a company's credit risk, often is not inherently interpretable.

Generation 3: Generative and Agentic AI Systems⁵ add stochastic behaviors and autonomy. LLMs⁶ deliver the capability to operate on vast multimodal data (text, audio, video, image) at scale, often approaching or exceeding expert-level performance.

Agentic systems allow for autonomous action under uncertainty to achieve a goal. Together, LLM-based agentic systems, though powerful, introduce stochasticity (in other words, the same input can yield different outputs), outsourced control (*for example*, if vendors like OpenAI are changing how a specific LLM works), and low transparency and interpretability.

Risks include behavioral drift (agents may take unanticipated paths to reach a goal), adversarial prompts, and goal misalignment—each underscoring the need for MRM to evolve.

EXAMPLE

A LLM agent operating for a bank, harnessed with access to unstructured context (for example, company financials, internal bank feeds, news articles), and a set of tools (for example, external APIs) can dynamically interpret the large data landscape. These agents can deliver real-time credit risk insights and can be used as part of an autonomous lending decision for small businesses, including narratives beyond traditional models' capabilities.

Each model generation comes with its own risk and its own risk mitigation techniques. Each generation expands model capabilities, from structured prediction to pattern recognition to reasoning and autonomous actions over unstructured data. But this pattern progressively sacrifices a portion of the transparency, determinism, and the controls on which traditional validation frameworks depend. As a result, Model Risk Management must fundamentally rethink its governance, its development, and its validation approaches.

³ Performance gains are incremental improvements; capability gains shift what is infeasible to feasible.

⁴ Modelers typically have full control of model inference hardware, as well as random seeds if applicable.

⁵ It is worth noting that not all Generation 3 models are equally complex and dynamic. Generation 3 systems range from autonomous agentic systems to LLM-augmented workflows, along with hybrids and intermediaries.

⁶ We adopt "LLM" as an umbrella term referring broadly to large generative models—including multimodal and diffusion-based architectures—not solely text-based language models.

3 Operationalizing Risk Mitigation

Managing model risk now demands continuous governance, not periodic validation. While frameworks (such as the SR 11-7) still serve as the gold standard for traditional models, their application must adapt to the complexity and opacity of AI systems. Institutions need dynamic controls embedded across the model lifecycle. Table 2 illustrates how risk mitigation strategies evolve across generations.

Table 2: Validation and Model Risk Management by Model Generation

	Gen 1: Traditional Models <i>The golden age of validation</i>	Gen 2: Machine Learning <i>The "black box"</i>	Gen 3: AI Systems <i>The "super black box"</i>
Key Risks	• Model misspecification and wrong functional form • Assumption and data-quality sensitivity (linearity, normality, missing data) • Parameter instability	• Overfitting and distribution shift • Spurious correlations and shortcut learning • Reduced transparency	• Stochastic, context-sensitive behavior (non-reproducibility, hallucination risk) • System opacity (complex action chains and orchestrations) • Vendor/dependency risk (closed models and unannounced updates)
Measures and Mitigations	• Classical statistical tests (specification, p-values, stationarity, backtests, out-of-sample tests) • Diagnostic analysis (residual plots, sensitivity / stability checks)	• Greater rigor in validation metrics (cross-validation, holdout performance, re-calibration) • Explainability and attribution (LIME, SHAP, perturbation / partial-dependence / feature importance) • Robustness checks (stress-tests, data-shift analysis)	• Behavioral tests (scenario evals, edge cases, adversarial tests / red-team, expert alignment quantification, optimality quantification) • Observability (monitoring, logging, tracing, drift and incident dashboards) • Operational controls (HITL oversight, autonomy control, guardrails and sandboxes)
Maturity			
Frequency			

Generation 1: Traditional models rely on established MRM practices. Deterministic models like logistic regression are governed through clear documentation, independent validation, stress testing, and periodic backtesting. Risks such as misspecification or parameter instability are traceable and manageable through classical statistical tests and diagnostic analysis.

EXAMPLE

Backtesting a credit risk scorecard against historical default data.

Generation 2: Explainability and robustness within deterministic boundaries. ML models remain deterministic; the same input yields the same output but also introduce opacity through non-linearities and large parameter sets. The following list is non-exhaustive. Mitigation strategies include:⁷

- Post-hoc explainability tools (*for example*, SHAP, LIME)⁸ to interpret predictions.
- Controlled deployment for specific, well-understood use cases.
- Human-in-the-loop oversight is essential for high-stakes decisions.

Periodic validation still works because model behavior is stable unless retrained. The challenge is interpretability, not unpredictability. Regulatory guidance is evolving to accommodate these methods, though practices are still working toward standardization across institutions.⁹

EXAMPLE

A credit risk Random Forest model uses SHAP values to show which features drive a borrower's credit risk.

Generation 3: Behavioral oversight and autonomy controls for non-deterministic systems. LLM-based agentic systems introduce stochastic outputs, autonomy, and vendor dependency, risks that invalidate static model validation cycles. Mitigation strategies must include:

- Behavioral testing through adversarial prompts (*for example*, what happens if altering a prompt sent to an agent); goal alignment checks to monitor the outcome of reasoning and autonomous decision-making.
- Continuous observability through real-time tracing and immutable audit trails of AI output and agentic steps (*for example*, tool calls or decision paths); implementation of dynamic constraints and guardrails, kill-switches (*for example*, if an agent takes clearly defined unintended actions), and escalation triggers (*for example*, what happens if unintended actions were taken?).
- Vendor risk (*for example*, from LLM providers) needs to be reduced by methods including fixing models to approved versions (version pinning) and maintaining backup models (with prompts/orchestrations) as fallback.

EXAMPLE

AI system passes all adversarial behavior tests ("red teaming"), aligns with credit risk expert expectations, and operates within predictable output variability. Deployment enforces monitoring and logging of all steps when generating real-time credit insights. Human-in-the-loop authorizes operation and maintains governance oversight.

Monitoring shifts from periodic to real-time and ongoing monitoring, as behavior can change due to vendor updates, emergent agent decisions, or simply model stochasticity. Governance must explicitly justify AI use due to its greater monitoring requirements.

⁷ A non-exhaustive list.

⁸ The SHapley Additive exPlanations (SHAP) method quantifies the contribution of each feature to a model's prediction by computing Shapley values, offering consistent and locally accurate explanations. The Local Interpretable Model-agnostic Explanations (LIME) method approximates complex models locally with interpretable ones (like linear models) to explain individual predictions by perturbing input data and observing changes in output.

⁹ See BIS (2025), Occasional Paper (No 24), "Managing explanations: how regulators can address AI explainability".

4 Case Study: Operationalizing MRM on AI Systems

Generation 3 systems may unlock a step-function change in capability, but the go/no-go decision hinges on whether MRM can be redesigned around accountability, enforceable control boundaries, and continuous evidence of behavior. Accordingly, we present a case study of Moody's Early Warning System (EWS) that operationalizes the Generation 3 governance controls in Table 2 that would not be addressed by traditional (SR 11-7) MRM frameworks.

Moody's EWS links traditional Generation 1 models for credit risk with Generation 2 models, news articles clustering algorithms, and with Generation 3 models, using LLMs to extract events from the news.

Specifically, the system converts news events into financial statement impacts and then feeds them downstream into Generation 1 credit risk models. The most critical steps are attributing firm relevant reports in news articles and clustering them into coherent events—areas where Generations 2 and 3 methods deliver a step change in accuracy, but also increase the need for measurable performance, accountable controls, and robust third party/API dependency management with defined fallback paths.

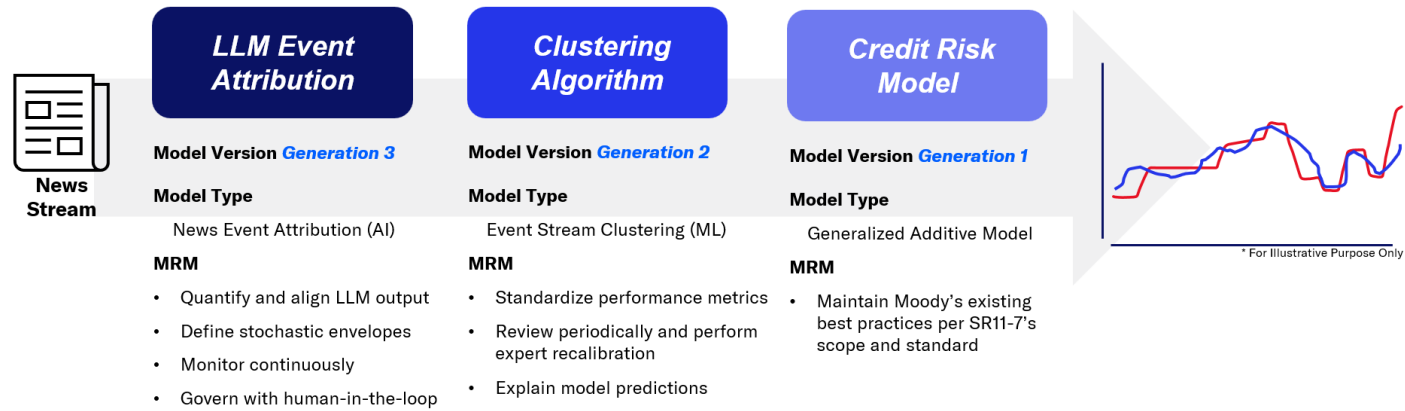


Figure 1: Moody's News-powered Early Warning System

The real-time EWS begins with an Attribution step using LLMs (Generation 3 model) that converts unstructured news articles into quantified firm-specific impacts. Because this Generation 3 component is non-deterministic and not inherently interpretable, we treat its performance as observable behavior that must be continuously monitored rather than assumed stable across validation cycles.

To ensure reliable performance, MRM combines expert review with quantitative measures. Specifically, we compare the AI's results to those of our internal subject matter experts using an expert-labeled dataset to assess how well the system recalls key statements. We also evaluate how closely the AI's rankings align with expert judgments and track

the variability in results across multiple runs. This process helps us define acceptable levels of randomness in the AI's output.

In production, structured output schemas and controlled sampling (such as adjusting temperature) help reduce variance. Continuous monitoring of extraction volumes, distributions, and variability provides ongoing evidence and triggers expert review (human-in-the-loop) when the system's behavior falls outside the validated range, capturing changes to the underlying models and unexpected behavior caused by their stochastic nature.

Changes to the prompt or to the model are governed through version control and revalidation, *for example*, a pipeline migrates from GPT-5-mini to a fine-tuned version of GPT-5-nano.

Downstream, a Clustering algorithm leveraging Generation 2 models (*for example*, K-means) deterministically groups incidence statements into coherent events. Using automated separation or tightness metrics (*for example*, Jaccard) complemented by periodic expert coherence review, we monitor cluster distributions over time with explicit revalidation triggers when upstream embeddings or material parameters change.

Moody's uses financial accounting models and general additive models (Generation 1) to translate news signals into credit risk estimates and early warning signals.¹⁰

As an integration with a Generation 1 model, validation focuses on the interface layer rather than revalidating core model mechanics already covered by Moody's existing MRM framework. Governance centers on input integrity, output reasonableness, and integration reliability; before invoking the credit risk model, inputs are validated for completeness, consistency, bounds, and format compliance. Outputs are checked for range and directional consistency (*for example*, alignment of credit risk movements with input changes). End-to-end integration and regression testing ensures consistent outputs for fixed inputs and establishes clear control boundaries, so the overall EWS remains governable end-to-end even when upstream components are partially stochastic.¹¹

This example demonstrates that significant capability gains from Generation 2, and Generation 3 can be realized without sacrificing end-to-end governability, provided controls are built around measured behavior, continuous monitoring, clear control boundaries, and human-in-the-loop steps. The result is a governance approach that is practical to operate, and that produces ongoing evidence suitable for internal oversight and supervisory expectations.

¹⁰ Detailed component metrics and triggers are provided in the appendices.

¹¹ Because the EWS operates as a predefined pipeline rather than an autonomous agent, this case study focuses on governing stochastic components within a bounded workflow. If the EWS outputs are used within an agentic decisioning layer, that layer should be governed under agentic-system MRM guidance.

5 Conclusion

Frameworks such as SR 11-7 remain relevant, but governance must evolve for AI. The three pillars of SR 11-7 (governance, development and implementation, and validation) still stand. What changes is the emphasis: governance must shift from static oversight to dynamic, continuous control. This includes setting dynamic guardrails instead of fixed thresholds, defining acceptance levels for third-party model risk, contingency plans, escalation triggers, and clear criteria for human-in-the-loop intervention.

Governance must justify the use of Machine Learning or Artificial Intelligence (such as agentic systems with large language models). The questions expand beyond "Is the model sound?" to include "Is AI justified here?", "What decisions will it inform?", and "Is the model's behavior stable?". As risks increasingly emerge from interactions between components and autonomous decision-making, monitoring must shift from isolated model elements to holistic system-level oversight.

The validator's role becomes more frequent, complex, and consequential. As modeling becomes commoditized, validators need new competencies. The same is true for decision makers playing a greater role in model governance. Prompt engineering, agentic context design, testing a system's behavior, observability infrastructure, and adversarial robustness assessment are skills that must be understood at various levels of an organization. Institutions must invest in upskilling teams to master these capabilities.

The pace of AI adoption will not slow; supervisory measures must respond. With undeniable performance gains and new capabilities, firms have strong incentives to adopt advanced models. Those firms that rebalance MRM toward explicit gatekeeping, scope of governance, and outcome-focused oversight, will deploy AI responsibly and will capture its benefits. Those that do not adopt these practices, risk either forgoing valuable capabilities or bearing risks they do not fully understand.

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